

## INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Electrical Engineering

2. Subject Name: Analog Electronic Circuits    L-T-P: 3-1-0    Credits: 4

3. Pre-requisites: None

4. Syllabus and reference books:

### Syllabus:

Introduction: Example of an electronic system, basic concept of signal, noise etc.

Semiconductor devices: Diode, BJT, MOSFET, their structures, principle of operations, electrical characteristics and their low frequency models

Amplifier models: Voltage amplifier, current amplifier, trans-conductance amplifier and trans-resistance amplifier

Biasing schemes for BJT and FET amplifiers, bias stability, various configurations (such as CE/CS, CB/CG, CC/CD) and their features

Small signal analysis: Estimation of voltage gain, input resistance, output resistance etc. for CE/CS, CB/CG, CC/CD amplifiers, and design procedure for particular specifications.

High frequency analysis: Miller's theorem, high frequency transistor models, frequency response of single stage amplifier (CE/CS, CB/CG, CC/CD amplifiers)

Current mirrors: Basic topology and its variants, V-I characteristics, output resistance and minimum sustainable voltage ( $V_{ON}$ ), maximum usable load, generation of current source.

Differential amplifier: Basic structure and principle of operation, calculation of differential gain, common-mode gain, CMRR and ICMR.

Feedback and Stability: Different feedback topologies such as voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., concept of stability.

Operational amplifier: Design of two-stage amplifier, frequency compensation, generalized structure of multistage amplifier.

Oscillators: Basic criteria for oscillation, Phase-shift, and Wien Bridge, oscillators, multivibrators (astable, monostable, bistable).

### Reference Books:

1. "Fundamentals of Microelectronics", B. Razavi, 2nd Edition.
2. "Microelectronic Circuits", A. S. Sedra and K. C. Smith, 7th Edition
3. "Analysis and Design of Analog Integrated Circuits", Gray, Hurst, Lewis, and Meyer, 5th Edition

## 5. Lecture-wise break-up:

| Sl. No.               | Topic                                                                                                                                                                                        | No. of lectures |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| 1.                    | Introduction: Example of an electronic system, basic concept of signal, noise etc.                                                                                                           | 1               |
| 2.                    | Semiconductor devices: Diode, BJT, MOSFET, their structures, principle of operations, electrical characteristics and their low frequency models                                              | 5               |
| 3.                    | Amplifier models: Voltage amplifier, current amplifier, trans-conductance amplifier and trans-resistance amplifier                                                                           | 1               |
| 4.                    | Biasing schemes for BJT and FET amplifiers, bias stability, various configurations (such as CE/CS, CB/CG, CC/CD) and their features                                                          | 5               |
| 5.                    | Small signal analysis: Estimation of voltage gain, input resistance, output resistance etc. for CE/CS, CB/CG, CC/CD amplifiers, and design procedure for particular specifications.          | 5               |
| 6.                    | High frequency analysis: Miller's theorem, high frequency transistor models, frequency response of single stage amplifier (CE/CS, CB/CG, CC/CD amplifiers)                                   | 5               |
| 7.                    | Current mirrors: Basic topology and its variants, V-I characteristics, output resistance and minimum sustainable voltage (VON), maximum usable load, generation of current source            | 2               |
| 8.                    | Differential amplifier: Basic structure and principle of operation, calculation of differential gain, common-mode gain, CMRR and ICMR.                                                       | 4               |
| 9.                    | Feedback and Stability: Different feedback topologies such as voltage series, current series, voltage shunt, current shunt, effect of feedback on gain, bandwidth etc., concept of stability | 5               |
| 10.                   | Operational amplifier: Design of two-stage amplifier, frequency compensation, generalized structure of multistage amplifier.                                                                 | 4               |
| 11.                   | Oscillators: Basic criteria for oscillation, Phase-shift, and Wien Bridge, oscillators, multivibrators (astable, monostable, bistable).                                                      | 3               |
| Total number of hours |                                                                                                                                                                                              | 40              |